

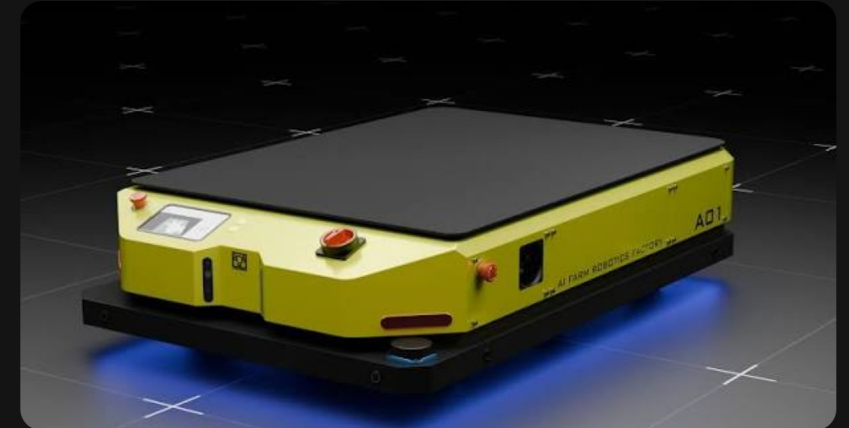
AI FARM ROBOTICS FACTORY · WEEK 1

COURSE: ROS 2 AMR & MANIPULATION

LINUX, MIDDLEWARE & ROS 2 ARCHITECTURE

Building the Software Foundation for Modern Robots

Ubuntu 24.04 · ROS 2 Jazzy · Gazebo Harmonic



Trailobot · differential-drive AMR

RB3-1200

6-DOF industrial manipulator

ABB IRB1200

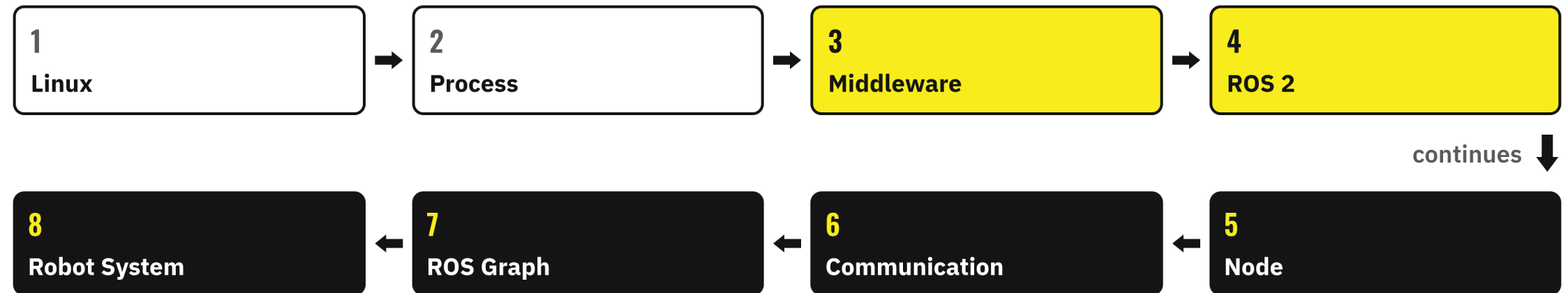
6-DOF industrial manipulator

ROS 2

Linux

WHAT WILL WE LEARN?

From Linux to a complete robot system

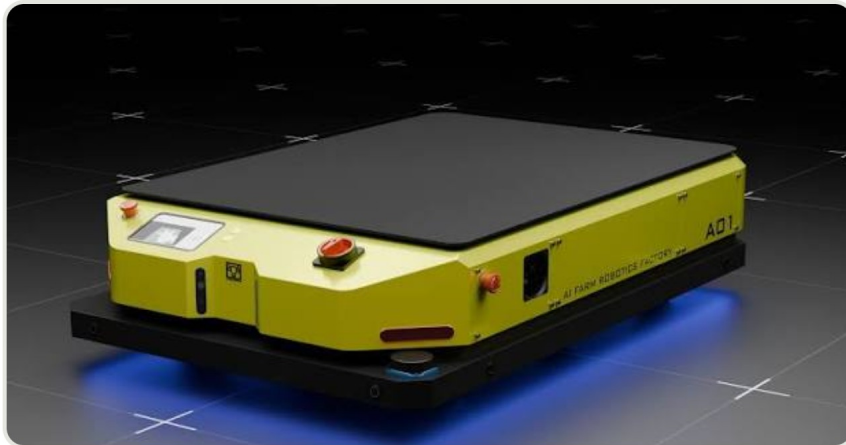


BY THE END OF THIS LECTURE, YOU SHOULD UNDERSTAND

Why Linux is widely used in robotics	What a process is	Why robots need middleware	What ROS 2 provides
What a ROS 2 node is	Topic / Service / Action / Parameter	How to inspect a ROS 2 system	How multiple computers can communicate

WHY DOES A ROBOT NEED SOFTWARE?

A robot is a combination of hardware + software.



Trailobot · differential-drive AMR

Sensors

LiDAR
IMU
Wheel encoders
Camera

Computer

Runs the software

Actuators

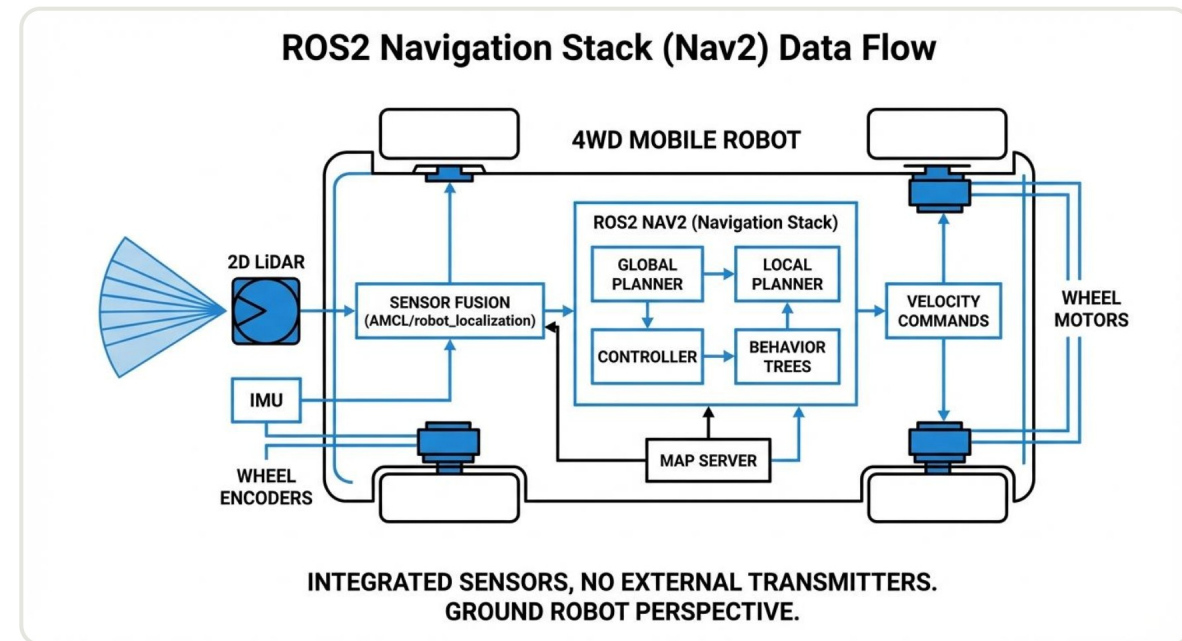
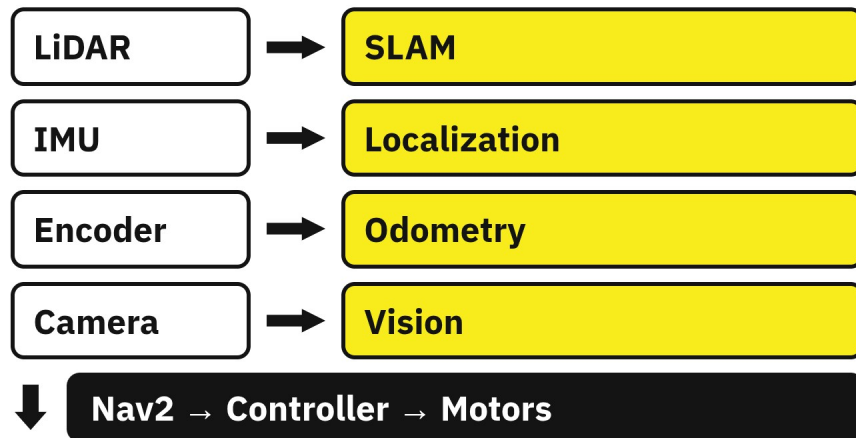
Motors
Wheels
Robot joints



SOFTWARE

THE ROBOT SOFTWARE PROBLEM

Many components must work together.



How can all these independent software components communicate reliably?

Answer: Middleware

FROM HARDWARE TO APPLICATION

Robotics software stack

ROBOT APPLICATION

SLAM / Nav2 / MoveIt

ROS 2

Nodes / Topics / Services

MIDDLEWARE / DDS

Communication layer

LINUX

Ubuntu 24.04

HARDWARE / NETWORK / GPU

Trailobot · RB3-1200 · ABB IRB1200



Services flow upward, from hardware to application.

KEY IDEA

Each layer provides services to the layer above it.

WHY LINUX FOR ROBOTICS?

WHY IS LINUX IMPORTANT?

Open-source OS

Powerful command line

Hardware and device access

Networking

Process management

Development tools

Large robotics ecosystem

ROS 2 support

COURSE ENVIRONMENT

UBUNTU 24.04 LTS

```
$ lsb_release -d
```

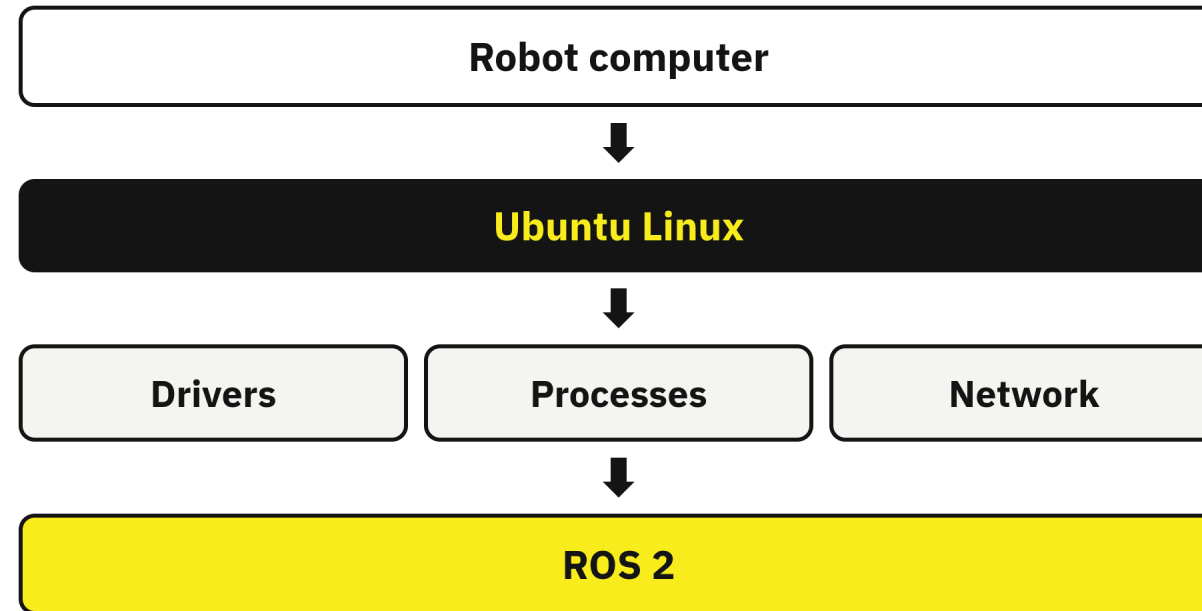
```
Description: Ubuntu 24.04 LTS
```

```
$ ros2 --help
```

Illustration of a terminal session. Replace with your own screenshot of the lab desktop.

LINUX IN A ROBOT

Linux is the foundation.



Your Trailobot computer can run:

LiDAR driver

Camera driver

ROS 2 nodes

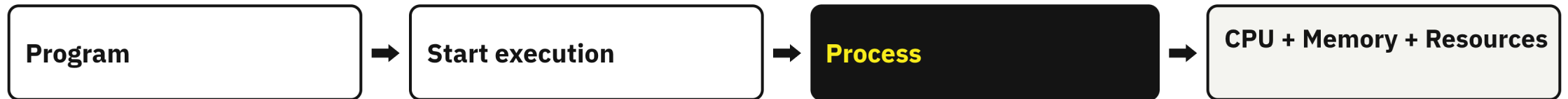
Navigation

Controllers

Diagnostic tools

WHAT IS A PROCESS?

A process = a running program



EXAMPLES ON A ROBOT

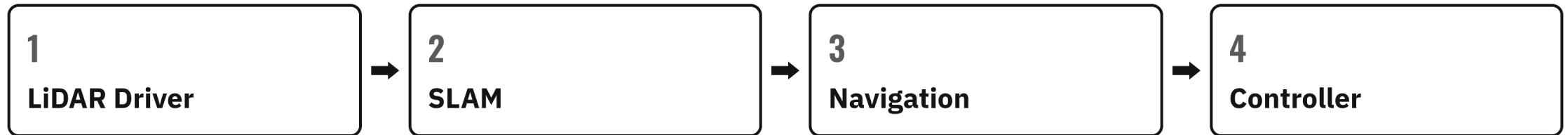
Software	Is a
LiDAR Driver	Process
SLAM	Process
Nav2	Process
Controller	Process
RViz2	Process

IMPORTANT

A ROS 2 node is commonly associated with a process, but not always one-to-one. ROS 2 also supports composable nodes.

MULTIPLE PROCESSES IN A ROBOT

A robot runs many programs at the same time.



Each box is a separate process

All four run at the same time and share the same computer (CPU, memory, network).

Data flows between them

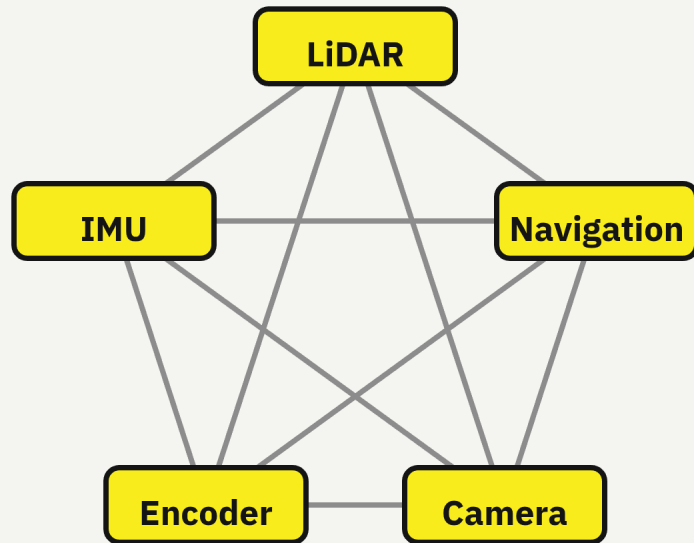
Scans, maps, plans and commands must travel from one process to another.

QUESTION

How should these programs communicate?

THE COMMUNICATION PROBLEM

Without a common communication system



As the system grows:

More connections

More complexity

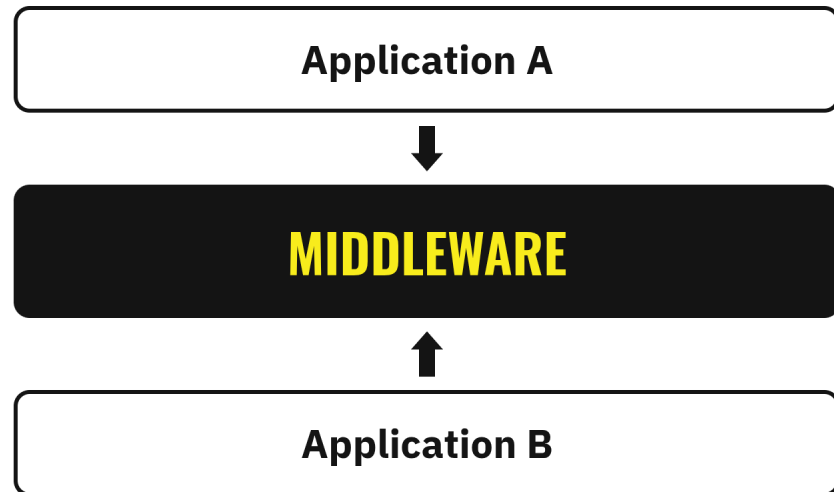
Harder debugging

Harder integration

Difficult distributed systems

WHAT IS MIDDLEWARE?

Middleware = communication software



Middleware sits between applications and the lower-level system. It can provide:

Communication

Data exchange

Discovery

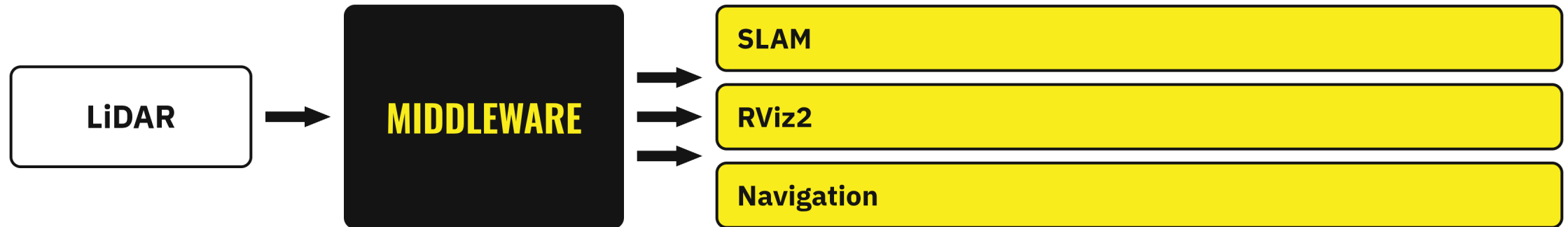
Serialization

Networking

Synchronization

WHY DO ROBOTS NEED MIDDLEWARE?

Robot example



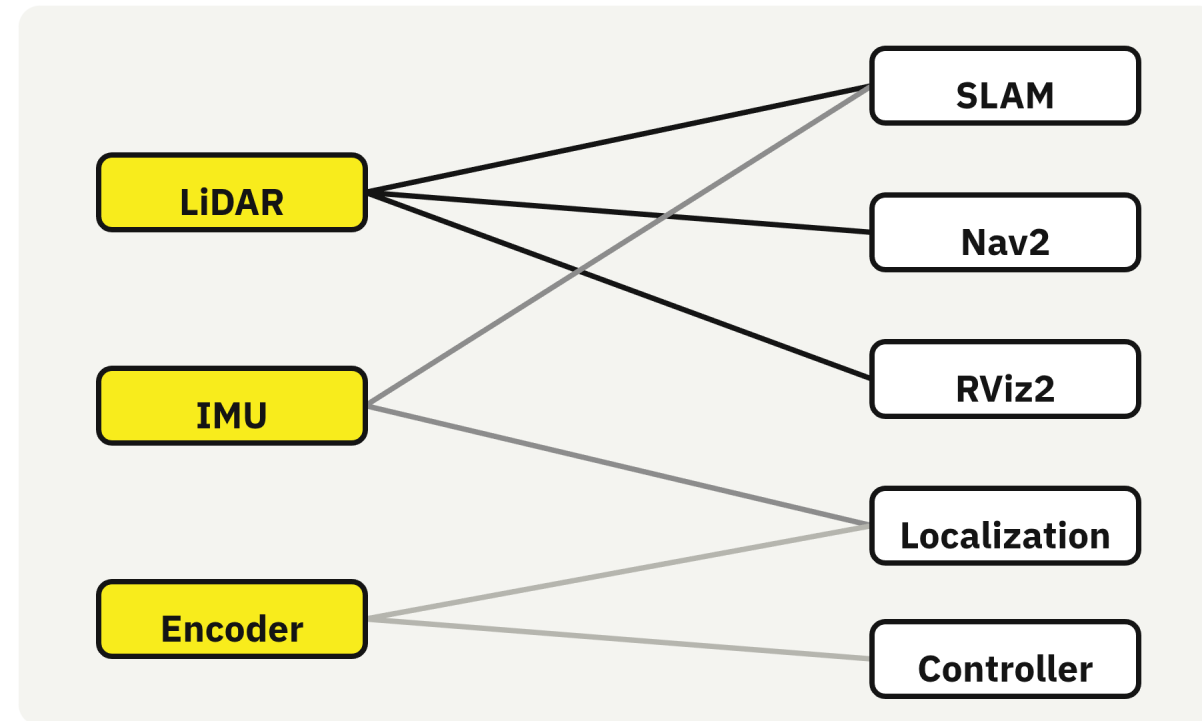
KEY IDEA

The LiDAR does not need to know every application that uses its data.

It only publishes data through the communication system.

ROBOT WITHOUT MIDDLEWARE

Direct communication



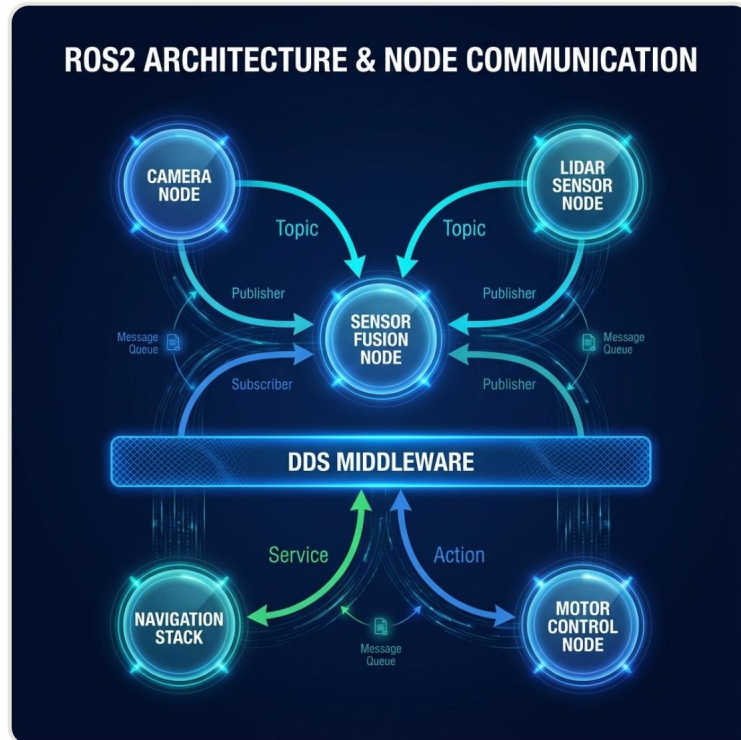
7

direct links for only 3 sensors and 5 programs

PROBLEM

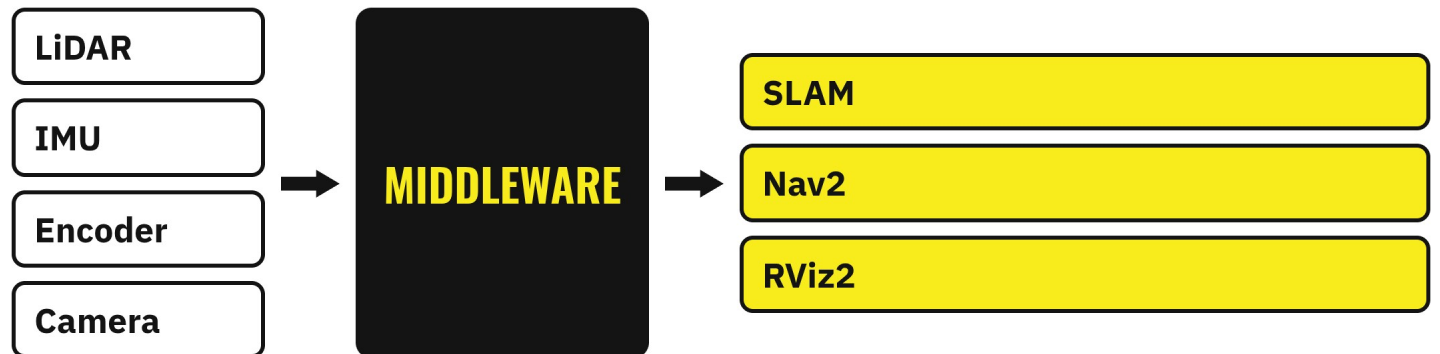
The number of connections grows quickly.

ROBOT WITH MIDDLEWARE



Nodes communicate through DDS middleware

Communication through a common system



MAIN BENEFIT

Components become more independent and easier to integrate.

WHAT IS ROS 2?

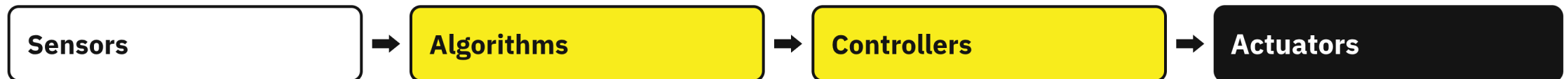
ROS 2 = Robot Operating System 2

BUT

ROS 2 IS NOT AN OPERATING SYSTEM.

ROS 2 is a robotics software framework and ecosystem. It provides tools, libraries, communication mechanisms and conventions for building robot applications.

IN THIS COURSE, ROS 2 CONNECTS



Reference: ros.org · docs.ros.org/en/jazzy

ROS 2 ARCHITECTURE

ROBOT APPLICATIONS

SLAM / Nav2 / MoveIt 2

ROS 2

Nodes / Topics / Services / Actions / Parameters

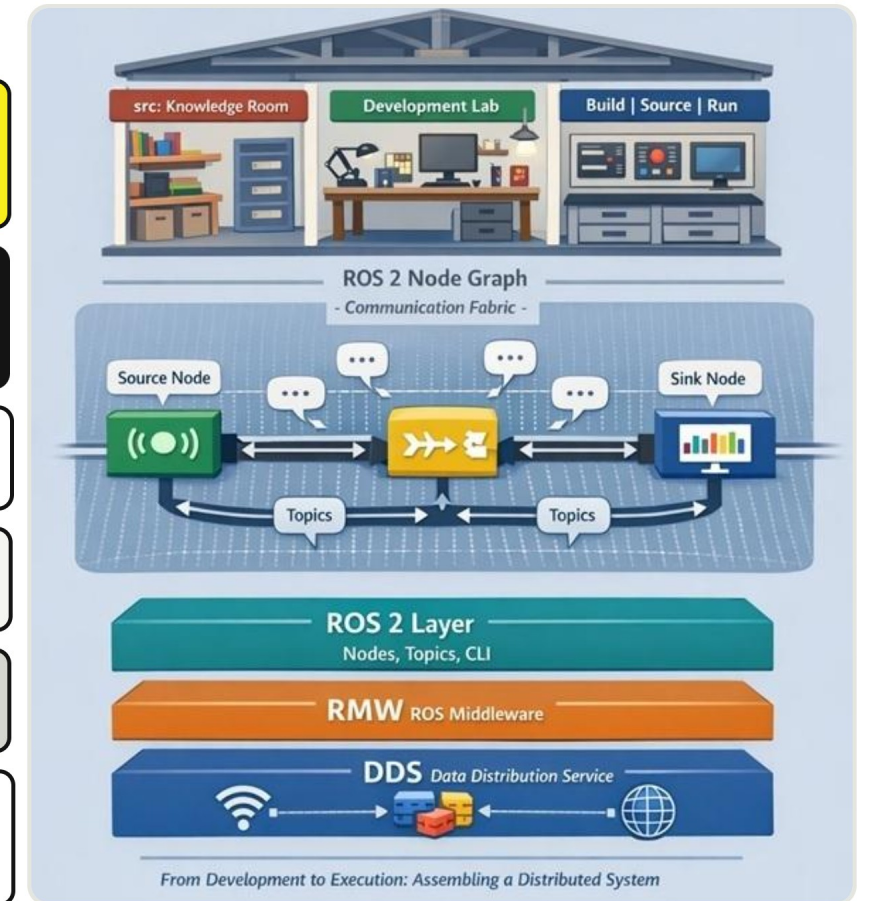
DDS MIDDLEWARE

UBUNTU LINUX

HARDWARE / NETWORK

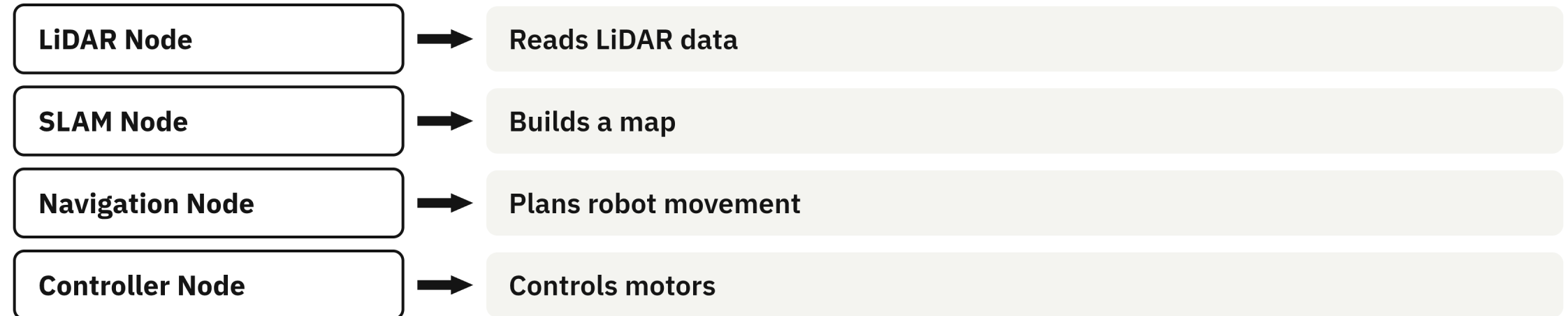
IMPORTANT

ROS 2 uses middleware underneath; commonly this is based on DDS implementations.



WHAT IS A ROS 2 NODE?

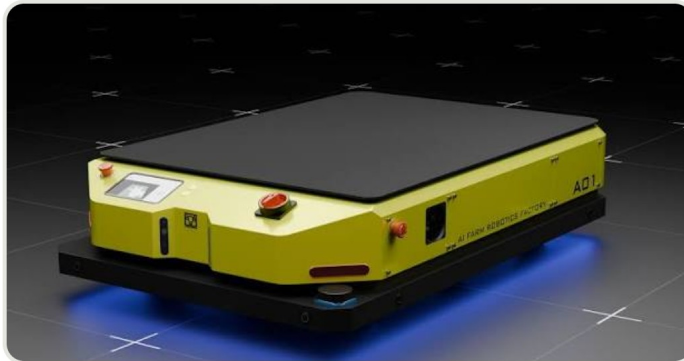
Node = a software component that normally performs a specific function



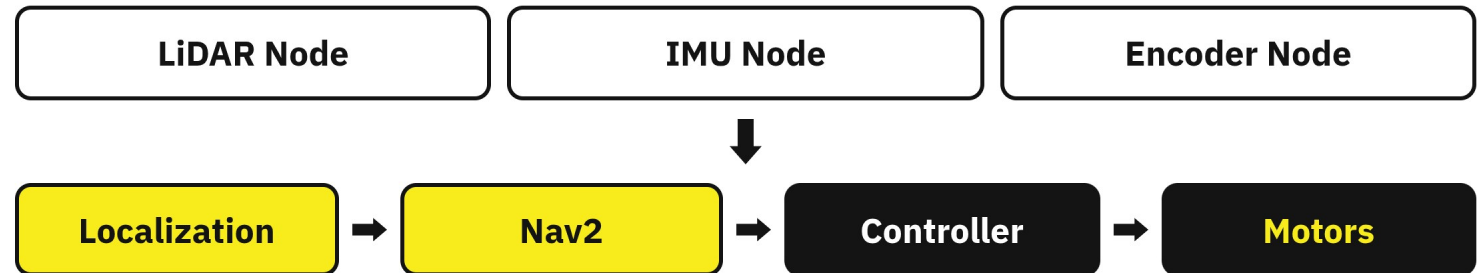
THINK OF A NODE AS
One useful software component inside the robot system.

A ROBOT MADE OF NODES

Trailobot example



Trailobot



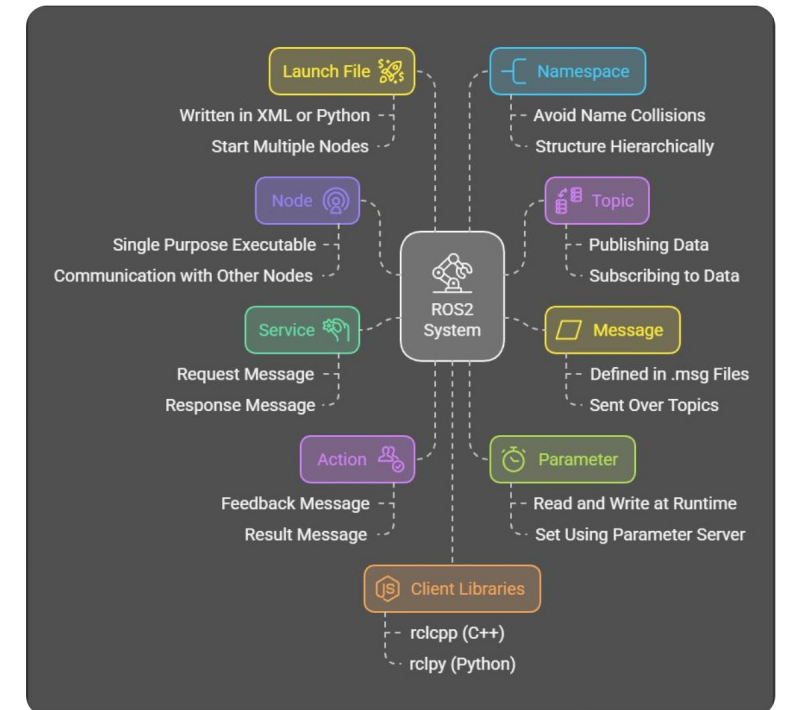
Each box is a node. Together they make the robot.

ROS 2 COMMUNICATION CONCEPTS

Five important concepts

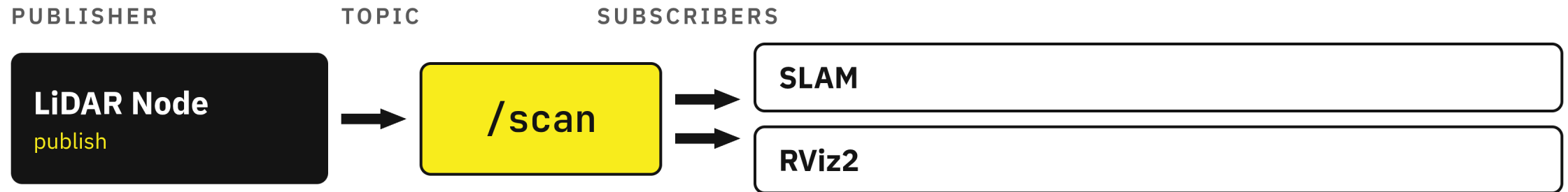
Concept	Purpose
Node	Software component
Topic	Continuous / asynchronous data
Service	Request → Response
Action	Long-running goal
Parameter	Node configuration

These five concepts are fundamental for the course.



TOPIC: PUBLISHER / SUBSCRIBER

Topic = data stream · Trailobot LiDAR example



IMPORTANT

The publisher does not need to know every subscriber.

GOOD FOR

LiDAR data

Camera images

IMU data

Odometry

Joint states

Velocity commands

COMMON ROBOT TOPICS

Example ROS 2 topics

TYPICAL MESSAGE TYPE

Topic	Message type
/scan	sensor_msgs/msg/LaserScan
/imu/data	sensor_msgs/msg/Imu
/odom	nav_msgs/msg/Odometry
/cmd_vel	geometry_msgs/msg/Twist
/joint_states	sensor_msgs/msg/JointState
/tf, /tf_static	tf2_msgs/msg/TFMessage
/camera/image_raw	sensor_msgs/msg/Image

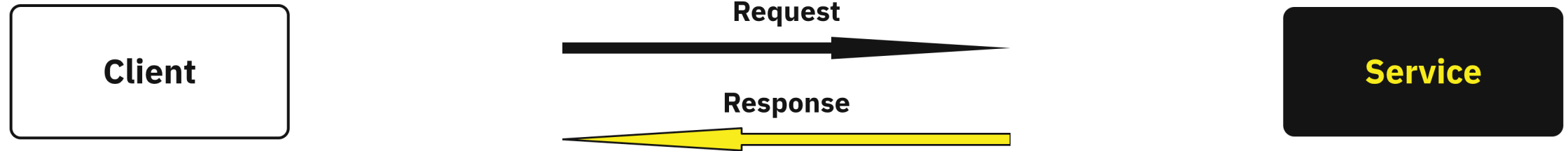
QUESTION

What kind of information would each topic carry?

Short classroom discussion.

SERVICE

Service = request → response

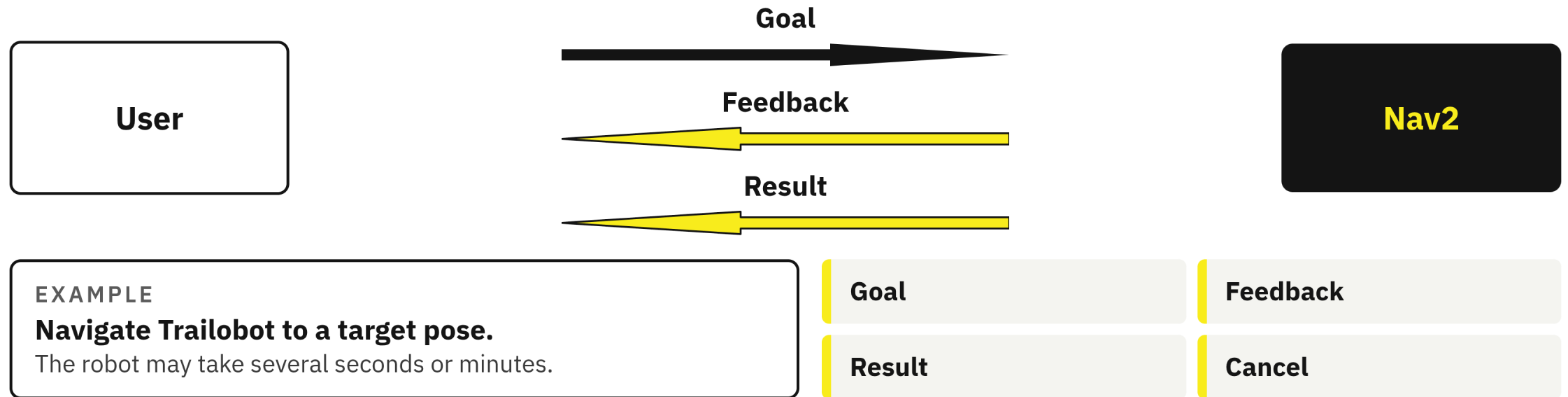


EXAMPLE



ACTION

Action = long-running task



TOPIC VS SERVICE VS ACTION

	Topic	Service	Action
Communication	Stream	Request / Response	Goal-based
Duration	Continuous	Short	Long-running
Feedback	Not built into pattern	Response	Yes
Example	/scan	/save_map	Navigate
Typical use	Sensor data	Quick command	Robot task

EASY WAY TO REMEMBER

Topic
 → **DATA**

Service
 → **REQUEST**

Action
 → **TASK**

PARAMETERS

Parameters configure node behavior

Node



speed

timeout

frame

```
max_velocity: 0.5
robot_radius: 0.25
use_sim_time: true
```

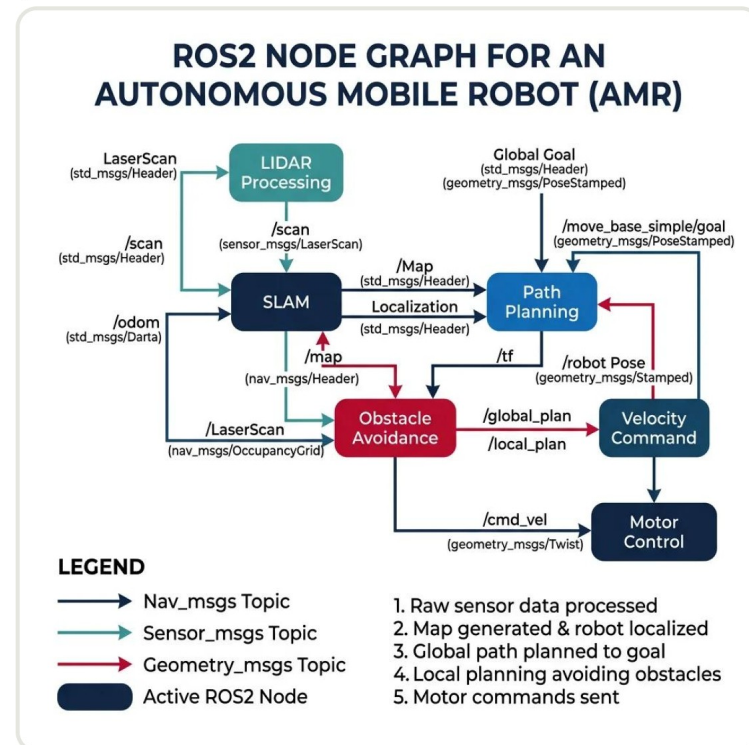
PARAMETER $\neq$ TOPIC

PARAMETER
Configures behavior.

TOPIC
Communicates runtime data.

ROS 2 GRAPH

ROS graph = nodes + communication relationships



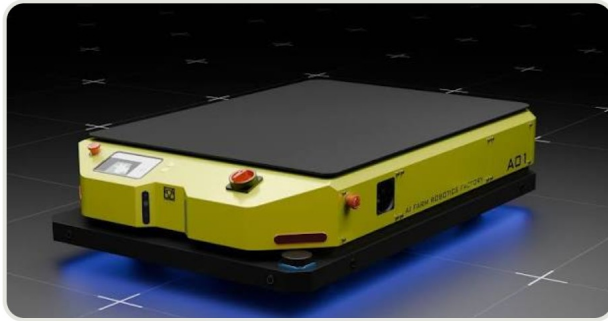
THE GRAPH HELPS US UNDERSTAND

Which nodes exist?

Which nodes communicate?

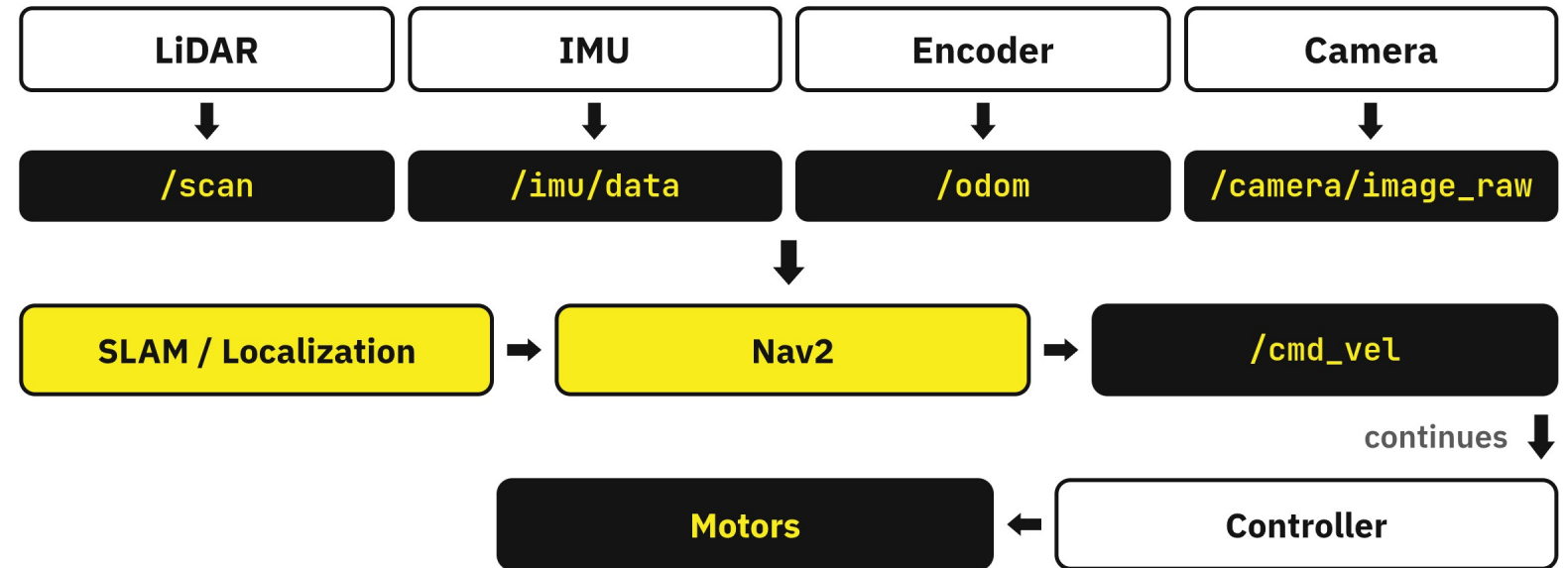
Which topics connect them?

COMPLETE TRAILOBOT ROS 2 SYSTEM



Course example: Trailobot

The most important integration slide of Week 1.



ROS 2 CLI

Command-line tools

```
ros2 node list
```

```
ros2 topic list
```

```
ros2 service list
```

```
ros2 action list
```

```
ros2 param list
```

WHY CLI?

The CLI allows us to inspect and debug a running robot system.

```
$ ros2 node list  
/lidar_node  
/slam_node  
/controller_node
```

Illustrative output. Replace with a screenshot of your own terminal.

INSPECT NODES

LIST NODES

```
ros2 node list
```

INSPECT A NODE

```
ros2 node info /node_name
```

WE CAN DISCOVER

Publishers

Subscribers

Services

Actions

```
$ ros2 node info /lidar_node
/lidar_node
Subscribers:
Publishers:
 /scan: sensor_msgs/msg/LaserScan
Service Servers:
Action Servers:
```

Illustrative output. Replace with a screenshot of real ROS 2 output.

INSPECT TOPICS

LIST TOPICS

```
ros2 topic list
```

INSPECT TOPIC

```
ros2 topic info /scan
```

READ DATA

```
ros2 topic echo /scan
```

```
$ ros2 topic info /scan
```

```
Type: sensor_msgs/msg/LaserScan
```

```
Publisher count: 1
```

```
Subscription count: 2
```

USEFUL FOR DEBUGGING

Is the sensor publishing data?

Illustrative output. Replace with a screenshot of the lab.

SERVICES AND ACTIONS

SERVICES

```
ros2 service list
```

Shows the quick request /
response commands.

```
/save_map
```

ACTIONS

```
ros2 action list
```

Shows the long-running tasks.

```
/navigate_to_pose
```

PARAMETERS

```
ros2 param list
```

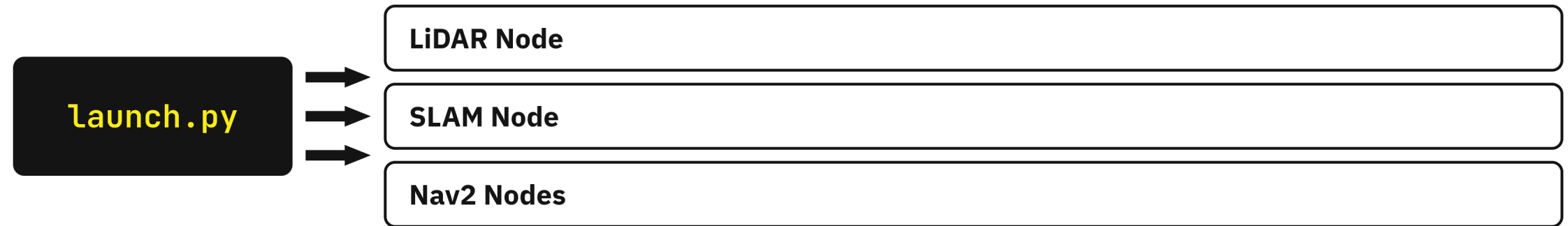
Shows the settings of each node.

```
use_sim_time
```

In the lab, run each command on the real environment and compare the output with the ROS graph.

LAUNCH SYSTEM

One launch file → multiple nodes



INSTEAD OF STARTING EVERYTHING MANUALLY

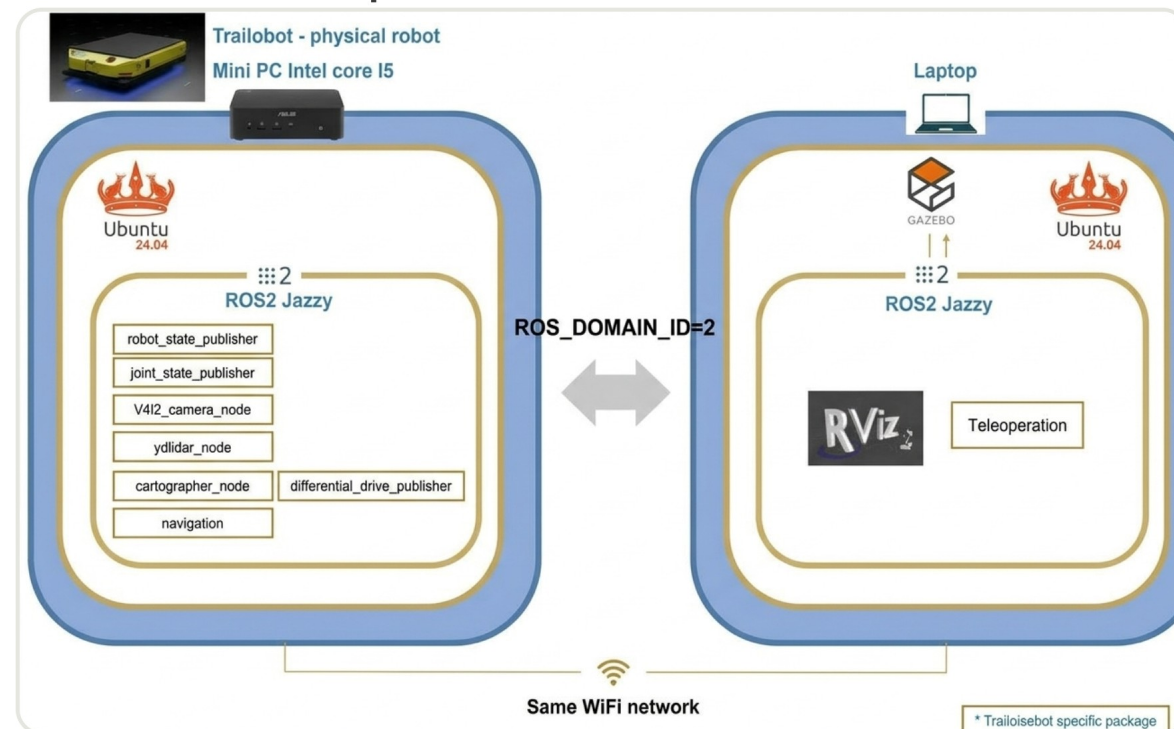
```
ros2 run ...  
ros2 run ...  
ros2 run ...  
ros2 run ...
```

WE CAN USE

```
ros2 launch <package> <launch_file>
```

DISTRIBUTED ROS 2

More than one computer



Example: a robot computer and a laptop on the same Wi-Fi network

Robot PC

Sensors · ROS 2 nodes · Controller

↑ Network ↓

Development PC

RViz2 · Debugging · Development

KEY IDEA

ROS 2 can support distributed robotic systems over a network.

KNOWLEDGE CHECK

1 What problem does middleware solve?

2 Is ROS 2 an operating system?

3 What is a ROS 2 node?

4 When should we use a topic?

5 What is the difference between a service and an action?

6 What is a parameter?

7 What does a ROS graph show?

KEY TAKEAWAYS

Remember these 7 ideas

1

Linux

Runs the software foundation.

2

Process

A running program.

3

Middleware

Helps software components communicate.

4

ROS 2

Robotics software framework / ecosystem.

5

Node

A software component.

6

Topic / Service / Action

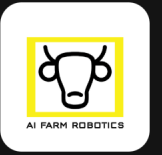
Different communication patterns.

7

ROS graph

Shows how robot software components connect.

**NEXT: BUILD IT IN
THE LAB**



NOW LET'S BUILD IT

LECTURE

Understand ROS 2



Understand nodes



Understand communication

LAB

1 Create workspace



2 Run ROS 2 nodes



3 Inspect topics

continues ↓

6 First ROS 2 system



5 Inspect actions



4 Inspect services

NEXT

Week 1 Lab · ROS 2 Architecture & Communication